

RAYAT BAHRA PROFESSIONAL UNIVERSITY



SEMESTER-I

Unit A -Web Design Principles and Web Page Design

Topic II- Working with Lists, Links, Images, Tables and Forms.

There are 3 different types of lists.



1.Ordered list

- Use the `<ol>` tag to begin an ordered list. Place the `<li>` (list item) tag between your opening `<ol>` and closing `</ol>` tags to create list items. Ordered simply means numbered

2.Unordered list

- Create a bulleted list with the `<ul>` tag. The bullet itself comes in three flavors: squares, discs, and circles. The default bullet displayed by most web browsers is the traditional full disc.

EXAMPLE OF UNORDERED LIST



```
<!DOCTYPE html>
<html>
<body>

<h2>An unordered HTML list</h2>

<ul>
  <li>Coffee</li>
  <li>Tea</li>
  <li>Milk</li>
</ul>

</body>
</html>
```

An unordered HTML list

- Coffee
- Tea
- Milk

type attribute



- This attribute controlled the visual marker used for each list item. Possible values included:
- disc (default): A filled circle.
- circle: A hollow circle.
- square: A filled square.
- none: No marker displayed.

type=Disc



```
<!DOCTYPE html>
<html>
<body>

<h2>Unordered List with Disc Bullets</h2>

<ul type="disc">
  <li>Coffee</li>
  <li>Tea</li>
  <li>Milk</li>
</ul>

</body>
</html>
```

Unordered List with Disc Bullets

- Coffee
- Tea
- Milk

type=Circle



```
<!DOCTYPE html>
<html>
<body>

<h2>Unordered List with Circle Bullets</h2>

<ul type="circle">
  <li>Coffee</li>
  <li>Tea</li>
  <li>Milk</li>
</ul>

</body>
</html>
```

Unordered List with Circle Bullets

- Coffee
- Tea
- Milk

Type=Square



```
<!DOCTYPE html>
<html>
<body>

<h2>Unordered List with Square Bullets</h2>

<ul type="square">
  <li>Coffee</li>
  <li>Tea</li>
  <li>Milk</li>
</ul>

</body>
</html>
```

Unordered List with Square Bullets

- Coffee
- Tea
- Milk

Type=none



```
<!DOCTYPE html>
<html>
<body>

<h2>Unordered List without Bullets</h2>

<ul type="none">
  <li>Coffee</li>
  <li>Tea</li>
  <li>Milk</li>
</ul>

</body>
</html>
```

Unordered List without Bullets

Coffee
Tea
Milk

EXAMPLE OF ORDERED LIST



```
<!DOCTYPE html>
<html>
<body>

<h2>An ordered HTML list</h2>

<ol>
  <li>Coffee</li>
  <li>Tea</li>
  <li>Milk</li>
</ol>

</body>
</html>
```

An ordered HTML list

1. Coffee
2. Tea
3. Milk

Ordered HTML List - The Type Attribute



The `type` attribute of the `<ol>` tag, defines the type of the list item marker:

Type	Description
<code>type="1"</code>	The list items will be numbered with numbers (default)
<code>type="A"</code>	The list items will be numbered with uppercase letters
<code>type="a"</code>	The list items will be numbered with lowercase letters
<code>type="I"</code>	The list items will be numbered with uppercase roman numbers
<code>type="i"</code>	The list items will be numbered with lowercase roman numbers

ORDERED LIST WITH UPPERCASE LETTERS



```
<!DOCTYPE html>
<html>
<body>

<h2>Ordered List with Letters</h2>

<ol type="A">
  <li>Coffee</li>
  <li>Tea</li>
  <li>Milk</li>
</ol>

</body>
</html>
```

Ordered List with Letters

- A. Coffee
- B. Tea
- C. Milk

HTML start Attribute



- The start attribute specifies the start value of the first list item in an ordered list.
- This value is always an integer, even when the numbering type is letters or romans. E.g., to start counting list items from the letter "c" or the roman number "iii", use start="3".
- Syntax
- `<ol start="number">`

 START ATTRIBUTE

```
<!DOCTYPE html>
<html>
<body>

<h1>The ol start attribute</h1>

<ol start="50">
  <li>Coffee</li>
  <li>Tea</li>
  <li>Milk</li>
</ol>

<ol type="I" start="50">
  <li>Coffee</li>
  <li>Tea</li>
  <li>Milk</li>
</ol>

</body>
</html>
|
```

The ol start attribute

50. Coffee
51. Tea
52. Milk

L. Coffee
LI. Tea
LII. Milk



3. Definition list

- Make definition lists as seen in dictionaries using the `<dl>` tag. These lists displace the term word just above the definition itself for a unique look. It's wise to bold the terms to displace them further.
- `<dl>` - defines the start of the list
- `<dt>` - definition term
- `<dd>` - defining definition

EXAMPLE OF DEFINATION LIST



```
<!DOCTYPE html>
<html>
<body>

<h2>A Description List</h2>

<dl>
  <dt>Coffee</dt>
  <dd>- black hot drink</dd>
  <dt>Milk</dt>
  <dd>- white cold drink</dd>
</dl>

</body>
</html>
```

A Description List

Coffee

- black hot drink

Milk

- white cold drink

CODE TO IMPLEMENT



```
<HTML>
<HEAD>
<TITLE>LISTS</TITLE>
</HEAD>
<BODY>
<H1>UNORDERED LIST</H1>
<UL TYPE="CIRCLE">
<LI>MANGO</LI>
<LI>ORANGE</LI>
<LI>APPLE</LI>
</UL>
<H1>ORDERED LIST</H1>
<OL TYPE=a START=3>
<LI>MANGO</LI>
<LI>ORANGE</LI>
<LI>APPLE</LI>
<H1>DEFINITION LIST</H1>
<DL>
<DT>KEYBOARD
<DD>AN INPUT DEVICE
<DT>PRINTER
<DD>AN OUTPUT DEVICE
</DL>
</BODY>
</HTML>
```

UNORDERED LIST

- MANGO
- ORANGE
- APPLE

ORDERED LIST

- c. MANGO
- d. ORANGE
- e. APPLE

DEFINITION LIST

KEYBOARD
AN INPUT DEVICE
PRINTER
AN OUTPUT DEVICE

HTML Links



- HTML Links - **Hyperlinks**
- HTML links are hyperlinks.
- You can click on a link and jump to another document.
- When you move the mouse over a link, the mouse arrow will turn into a little hand.
- **Syntax**
- The HTML **<a>** tag defines a hyperlink. It has the following syntax:
- `<a href="url">link text</a>`

EXAMPLE



```
<!DOCTYPE html>
<html>
<body>

<h1>HTML Links</h1>

<a href="https://www.GOOGLE.com/">Visit GOOGLE.COM!</a>

</body>
</html>
```

HTML Links

[Visit GOOGLE.COM!](https://www.GOOGLE.com/)

HTML LINKS CONTINUED...



- By default, links will appear as follows in all browsers:
- An unvisited link is **underlined and blue**
- A visited link is **underlined and purple**
- An active link is **underlined and red**

HTML Links - The target Attribute



By default, the linked page will be displayed in the current browser window. To change this, you must specify another target for the link.

The target attribute specifies where to open the linked document.

The target attribute can have one of the following values:

_self - Default. Opens the document in the same window/tab as it was clicked

_blank - Opens the document in a new window or tab

_parent - Opens the document in the parent frame

_top - Opens the document in the full body of the window

target="_blank"



```
<!DOCTYPE html>
<html>
<body>

<h2>The target Attribute</h2>

<a href="https://www.GOOGLE.com/" target="_blank">Visit GOOGLE!</a>

<p>If target="_blank", the link will open in a new browser window or tab.
</p>

</body>
</html>
```

The target Attribute

[Visit GOOGLE!](https://www.google.com/)

If target="_blank", the link will open in a new browser window or tab.

target="_self"



```
<!DOCTYPE html>
<html>
<body>

<h2>The target Attribute</h2>

<a href="https://www.GOOGLE.com/" target="_self">Visit GOOGLE!</a>

<p>_self - Default. Opens the document in the same window/tab as it was
clicked.</p>

</body>
</html>
```

The target Attribute

[Visit GOOGLE!](https://www.GOOGLE.com/)

_self - Default. Opens the document in the same window/tab as it was clicked.

Different Types of Links in HTML



- External links: External links are hyperlinks that connect one webpage to a page on a different website.
- Image links: Image links are hyperlinks that use images as the clickable element.
- Email links: Email links are hyperlinks that allow users to send an email directly from a webpage.
- Button as a Link

HTML Links - Use an Image as a Link



- To use an image as a link, just put the `<img>` tag inside the `<a>` tag.

```
<!DOCTYPE html>
<html>
<body>

<h2>Image as a Link</h2>

<p>The image below is a link. Try to click on it.</p>

<a href="page2.html"></a>

</body>
</html>
```

Image as a Link

The image below is a link. Try to click on it.



Link to an Email Address



- Use mailto: inside the href attribute to create a link that opens the user's email program

```
<!DOCTYPE html>
<html>
<body>

<h2>Link to an Email Address</h2>

<p>To create a link that opens in the user's email program</p>

<p><a href="mailto:rajveer216@gmail.com">Send email</a></p>

</body>
</html>
```

Link to an Email Address

To create a link that opens in the user's email program

[Send email](mailto:rajveer216@gmail.com)

Button as a Link



```
<html>
<body>
<h2>Button as a Links</h2>
<p>Click the button to go to the lists</p>
<button onclick="document.location='lists.html'
">Types of Lists</button>

</body>
</html>
```

Button as a Links

Click the button to go to the lists

Types of Lists

Images



- The HTML `<img>` tag is used to embed an image into a web page. It is an **empty element**, meaning it **does not** have a **closing tag**.
- The `<img>` tag is used to embed an image in an HTML page.
- **Notes:** Images are not technically inserted into a web page; images are linked to web pages.

The tag requires two essential attributes:



- The tag has two required attributes:

1. **src** - Specifies the path to the image

This attribute specifies the path or URL of the image file to be displayed. The path can be **relative** to the HTML document (e.g., images/my-image.jpg) or an **absolute** URL (e.g., <https://example.com/images/my-image.jpg>).

2. **alt** - Specifies an alternate text for the image, if the image for some reason cannot be displayed

Optional attributes that can be used with the `<img>` tag include:



- **width:** Specifies the width of the image in pixels or percentage.
- **height:** Specifies the height of the image in pixels or percentage.

EXAMPLE

```
<HTML>
```

```
<BODY>
```

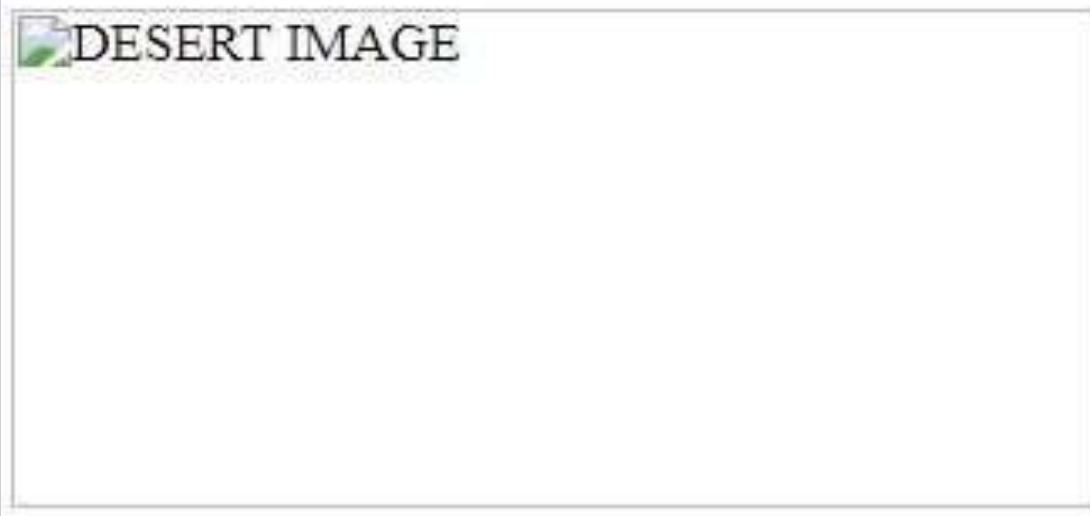
```
<IMG SRC="C:\Users\Public\Pictures\Sample Pictures\Desert.JPG" ALT="DESERT IMAGE"  
HEIGHT="50%" WIDTH="50%">
```

```
</BODY>
```

```
</HTML>
```



OUTPUT OF ALT TAG(if the image for some reason cannot be displayed)



TABLE



HTML tables are used to display tabular data, organizing information into rows and columns.

The fundamental elements for creating an HTML table are:

<table>: This is the main container element that defines the entire table.

<tr>: This element defines a **table row**. Each **<tr>** creates a new row in the table.

<th>: This element defines a **table header** cell. **<th>** elements are typically placed within the first **<tr>** to label the columns. By default, content within **<th>** is **bold** and **centered**.

<td>:

This element defines a standard **table data** cell. **<td>** elements hold the actual data for each cell within a row.

Commonly used HTML table attributes include:



1.border: Specifies the width of the border around the table and its cells in pixels.

Syntax

```
<table border="1">
```

2.align: Aligns the table horizontally on the page (e.g., "left", "center", "right")

```
<table align="center">
```



3. width: Sets the width of the table (in pixels or percentage).

```
<table width="50%">
```

4. height: Specifies the height of the table (in pixels or percentage)

```
<table height="50px">
```

5. colspan: Specifies the number of columns a cell should span.

```
<th colspan="2">Spanning two columns</td>
```



- **6. rowspan:** Specifies the number of rows a cell should span.

`<th rowspan="3">Spanning three rows</td>`

- 7. cellpadding:** Defines the space between the content of a cell and its border in pixels.

`<table cellpadding="5">`

- 8. cellspacing:** Specifies the space between individual table cells in pixels.

`<table cellspacing="2">`

Simple example



```
<table>
  <tr>
    <th>Name</th>
    <th>Class</th>
  </tr>
  <tr>
    <td>john</td>
    <td>CSE</td>
  </tr>
</table>
```

SIMPLE CODE



```
<html>
<body>
<table border="5" >
<tr><th>NAME</th><th>AGE</th></tr>
<tr ><td>shivi</td><td>20</td></tr>
<tr ><td>neha</td><td>30</td></tr>
</table>
</body>
</html>
```

A screenshot of a web browser displaying the rendered HTML table. The table has a thick black border and contains three rows of data. The first row has headers 'NAME' and 'AGE'. The second row has 'shivi' and '20'. The third row has 'neha' and '30'.

NAME	AGE
shivi	20
neha	30



```
<html>
```

```
<body>
```

```
<table align="center" border="5" width=50% height=50% >
```

```
<tr ><th rowspan=4>LIST</th> <th colspan=2>MARKS</th></tr>
```

```
<tr><th>NAME</th><th>AGE</th></tr>
```

```
<tr align="center"><td>shivi</td><td>20</td></tr>
```

```
<tr align="center"><td>neha</td><td>30</td></tr>
```

```
<caption align=bottom><b>Personal Information</b></caption>
```

```
</table>
```

```
</body>
```

```
</html>
```

OUTPUT



LIST	MARKS	
	NAME	AGE
	shivi	20
	neha	30

Personal Information

CODE WITH CELLSPACING AND CELLPADDING



```
<html>

<body>
<table border="5" width=50% height=50% cellpadding="50">

<tr><th rowspan=4>LIST</th> <th colspan=2>MARKS</th></tr>
<tr><th>NAME</th><th>AGE</th></tr>
<tr><td>shivi</td><td>20</td></tr>
<tr><td>neha</td><td>30</td></tr>
<caption align=bottom><b>Personal Information</b></caption>
</table>
</body>
</html>
```

OUTPUT



LIST	MARKS	
	NAME	AGE
	shivi	20
	neha	30

Personal Information

HTML FORM



- An HTML form is used to collect user input. The user input is most often sent to a server for processing.
- HTML forms, defined using the **<form> Tags** are essential for collecting user input on web pages.
- Over 85% of websites rely on forms to gather data from users, making them a fundamental component of modern web development.
- **Syntax:**
`<form>.....</form>`

The HTML **<input>** element is the most used form element.



- An **<input>** element can be displayed in many ways, depending on the type attribute.

Type	Description
<code><input type="text"></code>	Displays a single-line text input field
<code><input type="radio"></code>	Displays a radio button (for selecting one of many choices)
<code><input type="checkbox"></code>	Displays a checkbox (for selecting zero or more of many choices)
<code><input type="submit"></code>	Displays a submit button (for submitting the form)
<code><input type="button"></code>	Displays a clickable button

<input type="text">



```
<!DOCTYPE html>
<html>
<body>

<h2>Text input fields</h2>

<form>
  First name: <br> <input type="text"><br>
  Last name: <br> <input type="text"><br>
</form>

</body>
</html>
```

Text input fields

First name:

Last name:

<input type="radio">



```
<!DOCTYPE html>
<html>
<body>

<h2>Radio Buttons </h2>

<form>
  <input type="radio"> HTML <br>
  <input type="radio"> CSS<br>
</form>

</body>
</html>
```

Radio Buttons

- ☐ HTML
- ☐ CSS



```
<!DOCTYPE html>
<html>
<body>

<h2>Radio Buttons </h2>

<form>
  <input type="radio" name="radio"> HTML <br>
  <input type="radio" name="radio"> CSS<br>
</form>

</body>
</html>
```

Radio Buttons

- ☒ HTML
- ☐ CSS

<input type="checkbox">



```
<!DOCTYPE html>
<html>
<body>

<h2>Checkboxes to select multiple options</h2>

<form>
  <input type="checkbox"> I have a bike <br>
  <input type="checkbox"> I have a car<br>
  <input type="checkbox"> I have a boat<br>
</form>

</body>
</html>
```

Checkboxes to select multiple options

- ☐ I have a bike
- ☐ I have a car
- ☐ I have a boat

FORM EXAMPLE



```
<html>
<body>
<form height=50 width=50>
<pre>
<fieldset>
<legend align=center>INFORMATION
</legend>
First name:<input type="text">
Last name:<input type="text">
Password:<input type="Password">
D.O.B:<input type="datetime">
Qualification:<select><option>select course</option>
               <Option>B.tech</option>
               <option>nursing</option>
               </select>
Gender:<input type="radio" name=a >Male
<input type="radio" name=a>Female
```

CONTINUED...



About yourself

`<textarea rows=10 cols=100>Write 200 words</textarea>`

Select subject`<input type="checkbox">physics`

`<input type="checkbox">chemistry`

Select color: `<input type="color">`

Age: `<input type="number" min="18" max="60">`

Upload Resume:`<input type="file">`

`<input type="button" value="Sign In">`

`<input type="reset">`

`</fieldset>`

`</pre>`

`</form></body>`

`</html>`

OUTPUT

INFORMATION

First name:

Last name:

Password:

D.O.B:

Qualification:

Gender: ☐ Male

☐ Female

About yourself

Write 200 words

Select subject ☐ physics

☐ chemistry

Select color:



Age:

Upload Resume: No file chosen

<fieldset> and <legend>



- The <fieldset> element is a container used to group related elements within a web form.
- By default, browsers render a <fieldset> with a thin border around its content.
- This grouping is primarily for making forms more understandable for users.
- The <legend> element provides a caption or title for the <fieldset> it is placed within <fieldset> element.
- Browsers typically render the <legend> text embedded within the border of the <fieldset>.

Common HTML form controls include:



Text Input Controls:

<input type="text">: For single-line text input.

<input type="password">: For password input, obscuring the characters.

<textarea>: For multi-line text input.

Selection Controls:

<input type="radio">: For selecting a single option from a group.

<input type="checkbox">: For selecting zero or more options from a group.

<select>: For creating a drop-down list, often with **<option>** elements.



Button Controls:

`<input type="submit">`: For submitting the form data.

`<input type="reset">`: For resetting the form to its initial state.

`<input type="button">`: For a generic clickable button, typically used with JavaScript.

`<button>`: A more versatile button element that can contain content.



Specialized Input Types (Introduced in HTML5):

`<input type="email">`: For email addresses.

`<input type="url">`: For web addresses.

`<input type="number">`: For numeric input.

`<input type="range">`: For selecting a value within a specified range using a slider.

`<input type="date">`, `<input type="month">`, `<input type="week">`, `<input type="time">`, `<input type="datetime-local">`: For date and time input.

`<input type="file">`: For uploading files.



Hidden Controls:

`<input type="hidden">`: For storing data that needs to be submitted with the form but should not be visible to the user.

Other Controls:

`<datalist>`: Provides a list of pre-defined options for an `<input>` element with an associated list attribute.

`<output>`: Represents the result of a calculation.

`<progress>`: Displays the progress of a task.

`<meter>`: Represents a scalar measurement within a known range.

Example of datalist



```
<html>
<body>
<h1>The datalist element</h1>
```

```
<form>
```

Choose your browser from the list:

```
<input list="browsers" name="browser" id="browser">
```

```
<datalist id="browsers">
```

```
<option value="Edge">
```

```
<option value="Firefox">
```

```
<option value="Chrome">
```

```
<option value="Opera">
```

```
<option value="Safari">
```

```
</datalist>
```

```
<input type="submit">
```

```
</form>
```

```
</body>
```

```
</html>
```

The datalist element

Choose your browser from the list:

Edge

Firefox

Chrome

Opera

Safari

<meter> tag

The <meter> tag defines a scalar measurement within a known range.

```
<html>
```

```
<body>
```

```
<h1>The meter element</h1>
```

```
<p>The meter element is used to display a scalar  
measurement:</p>
```

```
Disk usage C:<meter value="2" min="0" max="10">2  
out of 10</meter><br>
```

```
Disk usage D:<meter value="0.6">60%</meter>
```

```
</body>
```

```
</html>
```

The meter element

The meter element is used to display a scalar measurement:

Disk usage C: 
Disk usage D: 